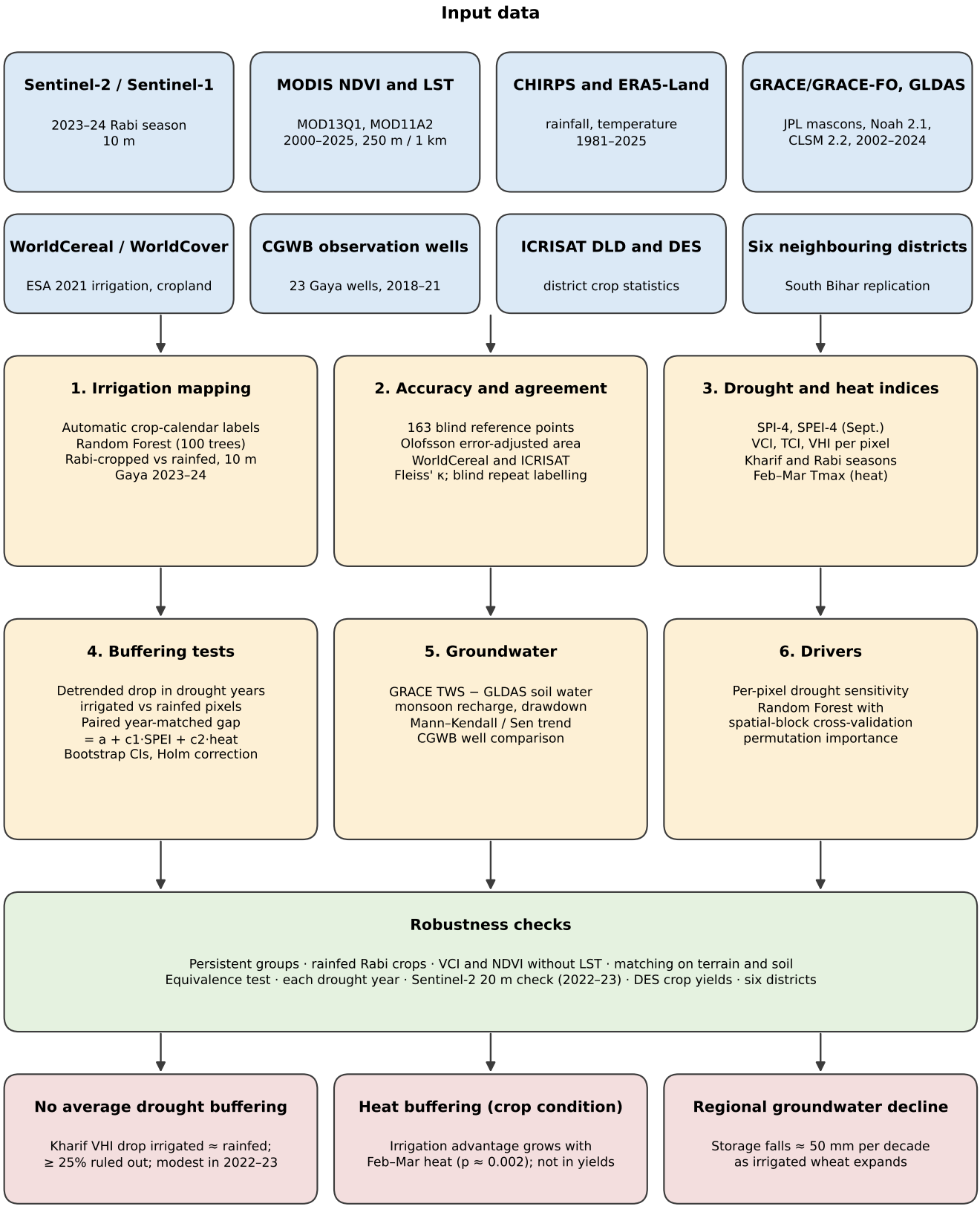
Input data
Sentinel-2 / Sentinel-1
2023–24 Rabi season
10 m
MODIS NDVI and LST
MOD13Q1, MOD11A2
2000–2025, 250 m / 1 km
CHIRPS and ERA5-Land
rainfall, temperature
1981–2025
GRACE/GRACE-FO, GLDAS
JPL mascons, Noah 2.1,
CLSM 2.2, 2002–2024
WorldCereal / WorldCover
ESA 2021 irrigation, cropland
CGWB observation wells
23 Gaya wells, 2018–21
ICRISAT DLD and DES
district crop statistics
Six neighbouring districts
South Bihar replication
1. Irrigation mapping
Automatic crop-calendar labels
Random Forest (100 trees)
Rabi-cropped vs rainfed, 10 m
Gaya 2023–24
2. Accuracy and agreement
163 blind reference points
Olofsson error-adjusted area
WorldCereal and ICRISAT
Fleiss' κ; blind repeat labelling
3. Drought and heat indices
SPI-4, SPEI-4 (Sept.)
VCI, TCI, VHI per pixel
Kharif and Rabi seasons
Feb–Mar Tmax (heat)
4. Buffering tests
Detrended drop in drought years
irrigated vs rainfed pixels
Paired year-matched gap
= a + c1·SPEI + c2·heat
Bootstrap CIs, Holm correction
5. Groundwater
GRACE TWS − GLDAS soil water
monsoon recharge, drawdown
Mann–Kendall / Sen trend
CGWB well comparison
6. Drivers
Per-pixel drought sensitivity
Random Forest with
spatial-block cross-validation
permutation importance
Robustness checks
Persistent groups · rainfed Rabi crops · VCI and NDVI without LST · matching on terrain and soil
Equivalence test · each drought year · Sentinel-2 20 m check (2022–23) · DES crop yields · six districts
No average drought buffering
Kharif VHI drop irrigated ≈ rainfed;
≥ 25% ruled out; modest in 2022–23
Heat buffering (crop condition)
Irrigation advantage grows with
Feb–Mar heat (p ≈ 0.002); not in yields
Regional groundwater decline
Storage falls ≈ 50 mm per decade
as irrigated wheat expands